

BhoomiShield — Product Requirement Document

Pan-India Cadastral GIS & AI Land Governance Platform (v3.0.0)

A unified, state-aware Digital Public Infrastructure connecting 28 State Revenue databases with deterministic 8-rule AI title verification, 3D Cadastral GIS, and grounded statutory legal intelligence.

Document Version: 3.0.0 (Production Release)

Compliance Standards: DILRMP, ULPIN (Bhu-Aadhaar), OGC WMS/WFS

Target Users: 1.4B Citizens, Buyers, Banks, Revenue Officers

Architecture Stack: React 18, Vite, FastAPI, SQLite/PostGIS, Leaflet

Security Standard: SHA-256 Checksums, DSC Sealing, Multi-Tier RBAC

Localization: 11 Regional Languages + 4 Accessibility Themes

1. Executive Summary & Product Vision

Land governance across India is fragmented across 28 States and 8 Union Territories, each maintaining disparate databases, terminology, and record formats (e.g., *Khatauni* in UP, *Satbara 7/12* in Maharashtra, *RTC Pahani* in Karnataka, *Khatian* in Jharkhand, *Patta* in Tamil Nadu). This fragmentation causes pervasive title uncertainty, multi-crore mortgage fraud in banks, delayed Right-to-Service mutation SLAs, and overwhelming civil litigation (which constitutes over **66% of all civil cases** in Indian courts).

BhoomiShield solves this structural breakdown by acting as a high-speed verification layer that unifies fragmented state databases into a single, canonical GIS map and deterministic AI risk engine—reducing title due diligence from **15–30 days down to under 5 seconds**.

Core Vision: To empower every Indian citizen, buyer, lending institution, and revenue official with tamper-proof, explainable, and instantaneous land records intelligence.

2. Problem Statement & Key Systemic Gaps

Structural Challenge	Ground Reality & Risk	BhoomiShield Automated Solution
Multi-Mortgage Loan Fraud	Borrowers take multiple loans against the same property across different bank branches using fake/duplicate deeds.	Real-time CERSAI + State Revenue lien sync; flags active mortgage charges instantly (Rule R006).
Chained Unmutated Deeds	Deeds registered at Sub-Registrar offices but never mutated in revenue registers (Register-II / Jamabandi).	Scans chain of custody; flags sellers whose names do not match official RoR ownership (Rule R001 & R004).
Statutory Tribal Land Violations	Illegal transfer of SC/ST or tribal land without mandatory District Magistrate sanction (e.g. UP Sec 98, MLRC Sec 36A).	Deterministic statutory restriction scanner; halts transactions lacking valid DM clearance (Rule R007).
Boundary & Area Discrepancies	Deed registered for 2.5 acres, but actual physical GIS survey measures only 2.1 acres (over-registration fraud).	Cadastral GIS polygon area calculation; flags variances > 5% between deed and ground map (Rule R002).
Mutation SLA Backlogs	Citizens waiting months for routine Dakhil-Kharij updates, breeding administrative friction.	Public 6-stage milestone tracker with automated 30-day statutory delay alerts (Rule R003).

3. User Personas & Access Control Matrix

User Role	Primary Workflows	Permitted Routes & Features	Security Gate
Citizen / Landowner	Verify land title, track pending mutations, calculate circle rates, store verified deeds in digital vault.	/search, /track-mutation, /valuation, /vault, /complaints	Public / OTP Access
Property Buyer & Investor	Pre-purchase risk checks, composite dispute audit, statutory restriction analysis, QR certificate validation.	/check-buy, /valuation, /legal-advisor, /verify	Public / Free Access
Bank / Lending Officer	Title clearance verification, hypothecation charge audit, cadastral polygon inspection for loan approval.	/check-buy, /verify, /valuation, /search	Institutional Login
Revenue Officer / Tahsildar	Mutation review queue, field report attachments, DSC digital sealing, cause list hearing updates.	/official (Dedicated Revenue Terminal)	High: Officer Challenge Key
Central Admin / Auditor	System telemetry, live database query studio, user role provisioning, tamper-proof audit trail ledger.	/admin (Master Database Console)	Critical: Root MFA Gate

4. Core Functional Modules & Detailed Specifications

1. National Portal Overview

Route: /

High-level national telemetry dashboard presenting total parcels indexed, active state integrations, and 1-click state demonstration loaders (UP, Maharashtra, Karnataka, Jharkhand, Bihar, Delhi).

2. Universal Land Search & Cadastral GIS

Route: /search

Multi-modal parcel query engine supporting dynamic state hierarchies (State → District → Tehsil → Village), state-specific field labels (*Khata/Gata/Survey* & *Khesra/Plot*), Ground GPS Pin-Drop search, and interactive Leaflet cadastral polygon rendering.

3. "Check Before You Buy" Pre-Purchase Title Scanner

Route: /check-buy

Automated title scanner that generates a **Composite Risk Score (0–100)** by simultaneously verifying owner concordance, boundary variances, mutation SLA aging, stay orders, and active bank hypothecations in < 5 seconds.

4. Live Mutation & SLA Tracker

Route: /track-mutation

Citizen-facing mutation tracking interface featuring an interactive 6-stage milestone timeline (*Submitted → Doc Verification → Field Inspection → Revenue Review → Final Decision → Record Update*) with automatic 30-day statutory delay badges.

5. Stamp Duty & Fair Valuation Calculator

Route: /valuation

Estimates fair market value vs. government circle rates with automatic multi-unit conversion (Sq. Ft, Acres, Bigha) and female buyer concession rebates (e.g., 1% stamp duty discount in UP and Delhi).

6. My Bhoomi Vault (Encrypted Locker)

Route: /vault

Encrypted personal document locker for registered deeds, Khatauni RoRs, and tax receipts with SHA-256 hash generation and digital verification request routing to local revenue officers.

7. State-Aware AI Legal Advisor

Route: /legal-advisor

Conversational statutory legal engine grounded in specific state land revenue codes (e.g., UP Revenue Code 2006, MLRC 1966, Karnataka PTCL 1978, CNT/SPT Acts) and central property acts (RERA 2016, Transfer of Property Act 1882).

8. Grievance Redressal Portal

Route: /complaints

Direct citizen grievance submission portal routing complaints to Tahsildars, SDM, and District Collectors with auto-generated unique complaint tracking IDs (IND-COMP-2026-XXXX).

9. Revenue Officer Workspace & DSC Sealing

Route: /official

Protected officer terminal for reviewing pending deed verification queues, applying cryptographic Digital Signature Certificate (DSC) seals, and scheduling Revenue Court cause list hearings.

10. Cryptographic Deed QR & Hash Verification

Route: /verify

Zero-trust verification tool for buyers, banks, and registrars to validate the integrity of issued BhoomiShield Land Verification Reports by checking SHA-256 hashes against the tamper-proof ledger.

11. Central Database Admin & SQLite Studio

Route: /admin

Protected administrative console for live inspection of 10,000+ database records, interactive SQL querying with execution safety limits, user role provisioning, and audit trail analytics.

12. Single Sign-On Gateway & Authentication

Route: /login

Role-based authentication gateway supporting Citizen, Officer, Banking, and Admin credentials with SHA-256 password hashing and session tokens.

5. Deterministic AI Risk Engine (Rules R001–R008)

Unlike opaque "black box" machine learning models, BhoomiShield uses a **deterministic, explainable rule engine** grounded in statutory revenue laws to calculate the Composite Land Risk Score:

Rule ID	Evaluation Check	Penalty	Statutory Grounding	Severity
R001	Owner Name Mismatch (RoR vs. Deed)	+25 Pts	Transfer of Property Act 1882, Sec 7	HIGH
R002	Boundary / Area Variance > 5%	+20 Pts	State Survey and Settlement Acts	MEDIUM
R003	Pending Mutation Overdue > 30 Days	+15 Pts	Right to Service (RTS) Act / Citizen Charter	MEDIUM
R004	Chained Prior Unmutated Deed Detected	+30 Pts	State Land Revenue Acts (Register-II / Jamabandi)	HIGH
R005	Active Court Stay Order or Lis Pendens	+35 Pts	Transfer of Property Act 1882, Sec 52	CRITICAL
R006	Active Bank Mortgage Hypothecation Charge	+20 Pts	SARFAESI Act 2002 / CERSAI Integration	MEDIUM
R007	SC/ST or Tribal Land Transfer Restriction	+40 Pts	UP Rev Code Sec 98 / MLRC Sec 36A / PTCL Sec 4	CRITICAL
R008	Non-Agricultural (NA) Land Use Misuse	+25 Pts	State Town & Country Planning / Revenue Acts	HIGH

Composite Risk Score Classification:

- **0 – 29: LOW RISK (Clear Title)** — Safe for purchase and bank mortgage disbursement.
- **30 – 59: MEDIUM RISK (Advisory Required)** — Requires resolution of minor survey or mutation delays.
- **60 – 79: HIGH RISK (Caution)** — Multiple discrepancies; requires written legal opinion.
- **80 – 100: CRITICAL RISK (Transaction Blocked)** — Severe statutory violations, active stay, or fraudulent seller.

6. Technical Architecture & Non-Functional Requirements

Architecture Layer	Technology Stack	Implementation Detail
Frontend Client	React 18 + TypeScript + Vite 5 + Tailwind CSS	Parallel manual chunking (5.9s build), responsive mobile-first layout.
Spatial GIS Engine	Leaflet 1.9 + GeoJSON Polygons	Dynamic cadastral polygon overlays with pan-zoom and area calculation.
Backend API	FastAPI (Python 3.11) + Uvicorn	Asynchronous REST endpoints with Pydantic v2 validation models.
Database	SQLite 3 (Demo/Prod) + PostgreSQL / PostGIS	Spatial index caching across 10,000+ parcels with foreign key constraints.
Security & Auth	SHA-256 Hashing + Challenge Gates + DSC	Role-Based Access Control isolating public, officer, and admin zones.
Localization & Themes	11 Indian Languages + 4 Dynamic Themes	English, Hindi, Bengali, Telugu, Marathi, Tamil, Urdu, Gujarati, Kannada, Malayalam, Odia.

7. Implementation Roadmap & Impact Metrics

Target Metric	Pre-BhoomiShield Baseline	BhoomiShield Target SLA	Impact Magnitude
Title Due-Diligence Time	15 – 30 Days (Manual Visits)	< 5 Seconds (Instant)	99.9% Time Reduction
Bank Mortgage Verification	7 – 14 Days (Lawyer Search)	Instant Automated Audit	Zero Mortgage Multi-Pledging
Search Latency	N/A (Disjointed Portals)	< 250 Milliseconds	Instantaneous Spatial Indexing
Citizen Linguistic Coverage	English / Hindi only	11 Regional Languages	1.4 Billion Citizens Covered